

employed four-pass printing, which meant that colors were applied to the paper individually, one after the other. As a result, printers were faster when printing black and white because with black and white printing, it is only necessary to apply the black. Thanks to technology advances, four-pass color laser printing is being supplanted by single-pass technology, in which all four colors are laid down simultaneously.

Now, it's possible to purchase color laser printers from HP that print up to 24 pages per minute in color. Inkjet printers have also benefited from advances in technology—models from HP now reach color printing speeds of 24 ppm as well. And some HP color multifunction devices reach color print speeds up to 22 ppm.

AFFORDABILITY

The best way to appreciate the financial value of color printing is to understand the value in terms of time to market, impact with customers, and total cost of ownership.

Eager to better manage their publishing expenses, Bellevue, Washington-based ArtSource, a digital media staffing and design agency, purchased multiple HP printers—including HP Color LaserJet 5500s, Color Inkjet cp1700s and Business Inkjet 1100s—that work together seamlessly to help increase productivity and customer satisfaction.

ArtSource founder Melody Christensen

estimates that the business marketing packet her company created using Microsoft® Publisher 2003 and printed in-house on HP printers costs less than \$5 per unit, but will generate eight new customers per month worth \$5,000 each. She further predicts that the annual revenues from this new source of income could

spend less time troubleshooting and fielding calls for help, and more time on strategic initiatives.

Reliability, as well as ease of use and interoperability of HP digital imaging solutions is helping the UK's BMW WilliamsF1 Team, a Formula One racing team, make a big impression with its Paddock Club

COLOR SUCCESS STORY

- **Company:** BMW WilliamsF1 Team, Oxfordshire, UK
- **Challenges:** Enhance guests' experiences of Formula One race weekends, provide a fast, easy-to-use solution for printing photographs for guests on race days.
- **Solution:** HP Photosmart digital cameras and printers.
- **Results:** Better customer relations, ease of use helps staff be more productive.

reach \$500,000 per year.

The bottom line is that professional-quality digital color is becoming more and more affordable. In fact, total cost of printer ownership (TCO) and cost per page have declined dramatically from the early days of color printing in the office.

RELIABILITY AND EASE OF USE

To their credit, monochrome laser printers from HP have developed stellar reputations for reliability and ease of use. The downside is that, in the past, this created unrealistic expectations for color laser printer performance. And unfortunately, many early color models failed to live up to those expectations.

According to Gartner, reliability has a large impact on printing costs, which is why it's so crucial in a business environment. The firm stated, "Reliability, quality and ease of use are key drivers of printing cost because they drive labor—the single biggest cost of printing."¹ User-required maintenance, end-user downtime and technical service were cited by Gartner as primary cost culprits.

Those culprits have become less culpable as color printer reliability has improved. Vendors such as HP have dedicated themselves to increasing reliability, which has led to significant improvements in the areas of design, manufacturing and testing. And more reliable devices help IT

guests on race weekends. Using HP Photosmart digital cameras and printers, the team's hospitality staff can take photos of the guests with their favorite drivers during tours of the pit lane, and then print those photos immediately, so guests may take autographed pictures home with them.

"This kind of initiative could not succeed without products that are extremely easy to use," said Jonathan Odell, sponsorship account manager for the BMW WilliamsF1 Team. "Hospitality is very important to the BMW WilliamsF1 Team. In order for guests to capture the moment, we use the innovative HP Photosmart solution. It is a great way of enhancing people's experience of Formula One racing as guests of the Paddock Club."

MAKING A DIFFERENCE WITH MEDIA

The enhanced quality of ink and media has also bolstered the image of color printers. Consider the opinion of David Axelbank, production manager at Magnum Photos, one of the world's largest global cooperatives of professional photographers, with offices in New York, London, Paris and Tokyo.

"As for the quality of ink on paper, in my opinion, the HP Designjet [printer] produces prints that very closely resemble traditional C-type prints. The colors were beautifully saturated and highly accurate."

Magnum used HP Designjet six-color printers to reproduce images for a recent series of international exhibitions. All prints were created using HP satin-finish photo paper and HP state-of-the-art inks designed for rich color depth, gloss uniformity, color stability and consistency.

